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LAB ASSIGNMENT-8

Border Gateway Protocol (BGP)

Title: Configuration and Analysis of BGP Routing

1. Objective

- To understand the working of **Border Gateway Protocol (BGP)**
- To configure **External BGP (eBGP)** between different Autonomous Systems
- To analyze **path selection using BGP attributes**

2. Theory

Border Gateway Protocol (BGP) is a **path vector routing protocol** used to exchange routing information between different **Autonomous Systems (AS)** on the Internet.

Key Concepts

- **Autonomous System (AS):** A group of networks under a single administrative domain
- **eBGP:** Routing between different AS
- **iBGP:** Routing within the same AS
- **Path Attributes:** Used to select best route
    - AS_PATH
    - NEXT_HOP
    - LOCAL_PREF
    - MED

BGP Characteristics

- Uses **TCP port 179**
- Supports **policy-based routing**
- Ensures **loop-free routing** using AS_PATH

3. Topology Description

Network Setup:

- 3 Routers (R1, R2, R3)
- Each router belongs to a different AS

- R1 → AS 100
- R2 → AS 200
- R3 → AS 300

- Each router connected to a LAN

4. IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.1.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/0	10.0.0.2	255.255.255.252
R2	G0/1	10.0.0.5	255.255.255.252
R3	G0/0	10.0.0.6	255.255.255.252
R3	G0/1	192.168.3.1	255.255.255.0

5. Procedure

Step 1: Create Topology

- Open Cisco Packet Tracer
- Place 3 routers and connect using serial/ethernet links
- Add PCs to each LAN

Step 2: Configure IP Addresses

Configure IP addresses on all router interfaces.

Example (R1):

```
conf t
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface g0/1
```

ip address 10.0.0.1 255.255.255.252

no shutdown

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip adress 192.168.1.1 255.255.255.0
                     ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#interface fa0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#router bgp 100
Router(config-router)#neighbor 10.0.0.2 remote-as 200
Router(config-router)#network 192.168.1.0 mask 255.255.255.0
Router(config-router)#
```

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Router1

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#interface fa0/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#router bgp 200
Router(config-router)#neighbor  10.0.0.1 remote-as 100
Router(config-router)#neighbor 10.0.0.6 remote-as 300
Router(config-router)#network 10.0.0.0 mask 255.255.255.252
Router(config-router)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface fa0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#router bgp 300
Router(config-router)#neighbor 10.0.0.5 remote-as 200
Router(config-router)#network 192.168.3.0 mask 255.255.255.0
Router(config-router)#
```

Step 3: Configure BGP

R1 Configuration (AS 100)

router bgp 100

neighbor 10.0.0.2 remote-as 200
network 192.168.1.0 mask 255.255.255.0

R2 Configuration (AS 200)

router bgp 200
neighbor 10.0.0.1 remote-as 100
neighbor 10.0.0.6 remote-as 300
network 10.0.0.0 mask 255.255.255.252

R3 Configuration (AS 300)

router bgp 300
neighbor 10.0.0.5 remote-as 200
network 192.168.3.0 mask 255.255.255.0

Step 4: Verify Configuration

Use the following commands:

show ip bgp
show ip route
show ip bgp summary

```
Router#show ip bgp summary
BGP router identifier 192.168.1.1, local AS number 100
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.2        4   200       0       0        1    0    0 00:16:12        4

Router#
```

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```
Router#show ip bgp
BGP table version is 1, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path

Router#
```

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```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, FastEthernet0/1
L       10.0.0.1/32 is directly connected, FastEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#
```

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